Hanshang Zhu

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Education

University College London (UCL) *2023 - 2027 (expected)*

MEng Robotics and Artificial Intelligence | **First-class average: 74%**

Relevant modules: Optimal Control, Robot Dynamics, SLAM, Deep Learning, Reinforcement Learning, Object-Oriented Programming, Linear Algebra, Scene Understanding and Awareness.

Research And Engineering Experience

Research Assistant, Humanoid Whole-Body Control | UCL *Jun 2026 - Present*

- Led Unitree G1 motion data processing; contributed to sim-to-sim experiments for the RA-L submission.
- Trained RL locomotion policies with a residual adapter for kinematic accuracy and style preservation; contributed to sim-to-real development and end-to-end real robot experiments.

World Action Model Distillation (WAM-OPD) | UCL *2026*

- Co-authored WAM-OPD (paper: **33.8% to 65.7%** success across 12 RoboTwin 2.0 tasks). Ran LingBot-VA to Flash-WAM on-policy distillation on ARX robot data; built offline open-loop evaluation using latent MSE and joint-space action L1.
- Built LingBot-VA FP8/W8A8/KV8 overlays on RTX 5090, targeting an emulated 24 GB budget; integrated InstinctFlash. Measured **13.9%** lower warm full-cycle latency vs the fastest measured BF16 setup; paired success difference vs the BF16 control **0.0 pp** (99 RoboTwin pairs, 95% CI +/-5.1 pp).

LingBot-VLA 2.0 Inference Acceleration *2026*

- Accelerated LingBot-VLA 2.0 on RTX 5090 (same 10 denoising steps): **1.6x** (p50) vs the official torch.compile server (**4.2x** vs eager), **35%** less steady GPU memory; paired RoboTwin success 82.1% to 81.2%. Used CUDA-graphed denoising, FP8 prefix linears and INT8 MoE experts on InstinctFlash.
- Identified and fixed a BF16 regression in the flow-matching time grid; upstream PR #59 is open.

Research Assistant, RoboHike | UCL *Mar 2026 - Present*

- Built a full-stack Unitree Go2 navigation system from SLAM (FAST-LIVO2) and Nav2 to locomotion control, including sim-to-real deployment and obstacle avoidance on hardware. Simulation project: UnrealNav.
- Developing semantic navigation with vision-language models (VLMs) for robust operation in the wild; building an onboard compute backpack powered by Jetson Thor.

Final-Year Project | UCL *2026 - 2027*

- Studying long-horizon WAM and VLA failures in legged mobile manipulation (Isaac Lab; quadruped + arm).

Agentic Engineering Intern | Cisco *Apr 2025 - Jun 2025*

- Built an evaluation pipeline for multi-agent LLM systems using LangChain and LangGraph.

Publications

1. L. Bao, **H. Zhu**, Y. Pan, et al., D. Kanoulas, C. Zhou. "Beyond Kinematic Tracking: Preserving Expressive Characteristics in Humanoid Whole-Body Control." *RA-L*, under review, 2026. Project page
2. L. Yang, Z. Zhong†, **H. Zhu†**, et al. "WAM-OPD: Joint Video-Action Supervision for World Action Model Post-Training with On-Policy Distillation." arXiv:2608.22364, 2026.

   Third author; †equal contribution with the second author.
3. L. Yang, **H. Zhu**, et al. "Reinforcement Learning Post-Training for Reasoning Large Language Models." *Unmanned Systems*, accepted, 2026. Survey.

Technical Skills

Robot learning: Diffusion and DiT policies with action chunking; on-policy distillation; DART; domain randomisation; world action models and VLAs.

Simulation and perception: ManiSkill3, SAPIEN, Isaac Lab, MuJoCo; RGB-D digital twins, 3D Gaussian splatting; FAST-LIO2 and FAST-LIVO2.

Hardware and tools: Unitree Go2, G1 and A2; RealSense; Jetson Orin and Thor; Python, PyTorch, ROS 2, Docker; Claude Code and Codex.